# CS3230 Lecture 6

## "Amortized Analysis"

NUS National University of Singapore
School of Computing

- Lecture Topics and Readings
  - Amortized Analysis [CLRS]-C17
    - *Introduction*
    - *Binary Increment*
    - *Dynamic Tables*

***It pays to analyze in detail.***
***Better analysis may come from***
***a different point of view***


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# CS3230 Mid-Term Quiz

**CS3230 Mid-Term Quiz**
Sat, 07-Mar, 10:00am – 11:30am
Venue: LT19 & SR3
(detail room assignment coming later)

**IMPORTANT:**

If you have VALID, GOOD reasons for a clash, email instructors ASAP giving details of your valid good reasons.


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# Sleator and Tarjan

Robert Tarjan and his students (most of them).
At a conference in 2008 to celebrate his 60th birthday.


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# Bob Tarjan's landmark paper…

- …one of many important papers…

*SIAM. J. on Algebraic and Discrete Methods*, 6(2), 306–318. (13 pages)

**Amortized Computational Complexity**

Robert Endre Tarjan

A powerful technique in the complexity analysis of data structures is amortization, or averaging over time. Amortized running time is a realistic but robust complexity measure for which we can obtain surprisingly tight upper and lower bounds on a variety of algorithms. By following the principle of designing algorithms whose amortized complexity is low, we obtain "self-adjusting" data structures that are simple, flexible and efficient. This paper surveys recent work by several researchers on amortized complexity.

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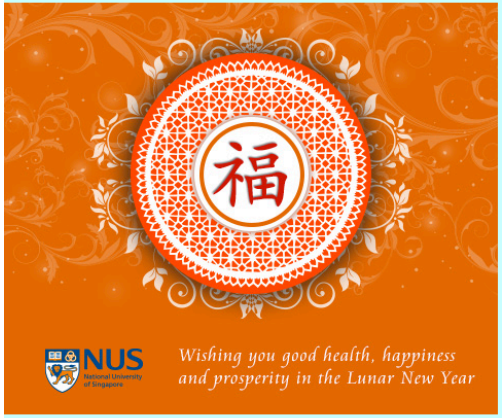
# CNY Greetings from CS3230 Staff




Dear All in CS3230,

Wishing you a Happy, Healthy and Rewarding Chinese New Year of the Goat, 2015 (you can still enjoy it, if you are not Chinese).

Have a good 1-week break. Come back from the break refreshed and ready for more CS3230 fun.

From: Hon Wai (and the CS3230 Staff)

As we celebrate NUS'110th year of founding in 2015, we invite you to click on the logo for more information on NUS110 celebrations.

NUS110 SG 50

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# Thank you.

# Q & A

NUS National University of Singapore
School of Computing


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# CS3230 Lecture 6 –

## "Amortized Analysis"

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### ❑ Lecture Topics and Readings

❖ Amortized Analysis [CLRS]-C17

- ◆ *Introduction*
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***It pays to analyze in detail.***
***Better analysis may come from***
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# Data Structure Operations

Given *any* sequence of $n$ operations $O_1, O_2, \ldots, O_n$ on a data structure $D$ where
$D_k$ = d.s. after applying $O_k$ with cost $c_k$ on $D_{k-1}$.

$$D_0 \xrightarrow{O_1} D_1 \xrightarrow{O_2} D_1 \xrightarrow{O_3}$$
$$\cdots \xrightarrow{O_{k-1}} D_{k-1} \xrightarrow{O_k} D_k$$

**Want to compute $C(n)/n$, where**
$$C(n)/n = (c_1 + c_2 + c_3 + \ldots + c_n) / n = \frac{1}{n}\sum_{k=1}^{n} c_k$$


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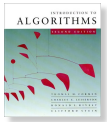
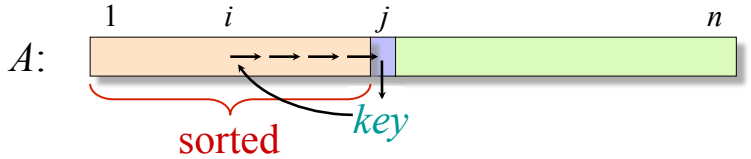


# Consider: Insertion sort

```
INSERTION-SORT (A, n)        ▷ A[1 . . n]
  for j ← 2 to n
      do key ← A[j]
         i ← j – 1
         while i > 0 and A[i] > key
             do A[i+1] ← A[i]
                i ← i – 1
         A[i+1] = key
```

$O_j$ = "inserts" $A[j]$ into sorted $A[1 .. (j–1)]$

operation $O_j$ with cost is $c_j$






# Insertion Sort: $(n–1)$ "insert" ops

Insertion Sort does $(n–1)$ "insert" operations

Let $O_j$ = "insert" $A[j]$ into sorted $A[1 .. (j–1)]$

Let $c_j$ = the cost of operation $O_j$

Consider a worst-case "insert" sequence:

In the **worst-case**, $c_j = (j–1)$

*Total* (*worst-case*) *time* for Insertion Sort:

$C(n) = (c_1 + c_2 + c_3 + \ldots + c_n)$

$= (0 + 1 + 2 + \ldots + (n–1)) \approx \mathbf{0.5(n^2)}$

Average cost of "insert" op. (**worst-case**):

$C(n)/n \approx \mathbf{0.5n}$

This worst-case bound is **TIGHT**.
Eg: when $A[1..n]$ reversed-sorted.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# Insert into Binary Search Tree

Do $n$ INSERT ops on an *initially empty* BST

$j^{th}$ INSERT operation inserts into BST (of size $(j–1)$)

Let $d_j$ be the cost of $j^{th}$ INSERT operation

Consider a worst-case INSERT sequence

In the worst-case, $d_j = (j–1)$

*Total* (worst-case) *time* for $n$ INSERT's:

$C(n) = (d_1 + d_2 + d_3 + \ldots + d_n)$

$= (0 + 1 + 2 + \ldots + (n–1)) \approx \mathbf{0.5(n^2)}$

Average cost of INSERT op. (worst-case):

$C(n)/n \approx \mathbf{0.5n}$

This worst-case bound is **TIGHT**.
Eg: elements inserted into BST in increasing order.


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# Motivation for Amortized C

For Insertion Sort, Insert into BST,

- ❖ worst-case upper-bound is **TIGHT**
- ❖ Achieved when array is reversed-sorted and when elements inserted in increasing order.

*But*, sometimes,

- ❖ simple worst-case upper-bound is **NOT TIGHT**
- ❖ May *over-estimate* the *actual* running time


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# Example: Consider this sequence

**Consider a sequence of operations with cost**

| $k$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $c_k$ | | | | | | | | | | |

| $k$ | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|
| $c_k$ | | | | | | | | | | |

$$c_k = \begin{cases} k & \text{if } (k-1) \text{ is a power of 2} \\ 1 & \text{otherwise} \end{cases}$$


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# Example: Consider this sequence

**Consider a sequence of operations with cost**

| $k$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $c_k$ | 1 | 2 | 3 | 1 | 5 | 1 | 1 | 1 | 9 | 1 |

| $k$ | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|
| $c_k$ | 1 | 1 | 1 | 1 | 1 | 1 | 17 | 1 | 1 | 1 |

$$c_k = \begin{cases} k & \text{if } (k-1) \text{ is a power of 2} \\ 1 & \text{otherwise} \end{cases}$$


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# Amortized Complexity A&E

**Question:**
How many of you keep a "Clean and Tidy Room" ***all the time***?

Let $CTR_k$ = "CLEAN-AND-TIDY-ROOM" op on the $k^{th}$ day.

Yes, we know that $CTR_k$ is "expensive" operation!

**Answer:**
Most of us a lazy most of the time.
But *once in a while*, *must do a lot of cleanup*


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# Amortized Complexity A&E

**Su Yuen Chin**
19 hrs · Singapore ·

This just happened. Me wake up, all blurry-eyed, chill in bed... open phone, see whatsapp. Message from mum? Oh ok. Open message

"Hi dear, r u home? I'm on way to Singapore and will be arriving at 5pm"

*pinches face and stares at message again* OMG!! THIS IS REAL!!!

*instantly transforms into high speed cleaning robot!*

Like · Comment · Share

Eugene Wee and 23 others like this.

On Sat, 13-9-2014 this appeared on my FB wall.



# Amortized Complexity A&E

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Like · Comment · Share

Eugene Wee and 23 others like this.

**Henry Bai Shuyong** Why must clean when your mother come?
19 hrs · Like · 1

**Su Yuen Chin** Henry Bai Shuyong Ya hor, my place so clean already right! unfortunately still fails my mum's standards
19 hrs · Edited · Like · 2

**Hon Wai Leong** SuYuen;

This is so funny, yet so common.
Can I use this for class? When I teach amortized complexity analysis...

--honwai
19 hrs · Like · 4

On Sat, 13-9-2014 this appeared on my FB wall.

Me think… Amortized Complexity in real life!!



# Amortized Complexity A&E

**Su Yuen Chin** Hon Wai Leong Sure thing! How is this related to that though? o_O?
21 hrs · Like

**Hon Wai Leong** Long answer:

In optimizing for amortized time efficiency, we get lazy with data structure updates when we don't have to properly organize things. So, we are lazy (but fast) most of the time.

But when we need to, we just work much harder and we will have to get EVERYTHING or MOST things back in order. So, once in a while (like parent visit, in-laws visit), we work like crazy hard. HAHA, like you now. Sorry to pour salt on wound, wicked hehehe.

But overall, worst-case performance is still better than maintaining order each and every time.

So, back to your question:
Like how many student/people keep their rooms/house/office clean and neat ALL the time?

No, most people don't.
We are lazy most times, and we work hard and get things organized when we have to.

So, most students practice amortized complexity optimization!

--honwai
20 hrs · Like · 13

All students practise amortized complexity optimization



# Example: Consider this sequence

## Consider a sequence of operations with cost

| $k$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $c_k$ | 1 | 2 | 3 | 1 | 5 | 1 | 1 | 1 | 9 | 1 |

| $k$ | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|
| $c_k$ | 1 | 1 | 1 | 1 | 1 | 1 | 17 | 1 | 1 | 1 |

$$c_k = \begin{cases} k & \text{if } (k-1) \text{ is a power of 2} \\ 1 & \text{otherwise} \end{cases}$$


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# When does worst-case happen?

**Consider a sequence of operations with cost**

$$c_k = \begin{cases} k & \text{if } (k-1) \text{ is a power of 2} \\ 1 & \text{otherwise} \end{cases}$$

***Worst case happens here***

Most operations only cost 1

**Worst-case: (Upper bound $W_k$)**
$c_k \le k = W_k$ for all $k$

**Worst case** is *expensive* (but not often)
most operations are cheap (cost=1)


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# If we bound each operation by $W_k$

**Consider a sequence of operations with cost**

| $k$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $c_k$ | 1 | 2 | 3 | 1 | 5 | 1 | 1 | 1 | 9 | 1 |
| $W_k$ | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **10** |

| $k$ | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|
| $c_k$ | 1 | 1 | 1 | 1 | 1 | 1 | 17 | 1 | 1 | 1 |
| $W_k$ | **11** | **12** | **13** | **14** | **15** | **16** | **17** | **18** | **19** | **20** |


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# If we use worst-case bound

**Consider a sequence of operations with cost**

$$c_k = \begin{cases} k & \text{if } (k-1) \text{ is a power of 2} \\ 1 & \text{otherwise} \end{cases}$$

**Worst-case: (Upper bound $W_k$)**
$c_k \le k = W_k$ for all $k$

Total time for $n$ operations
$C(n) \le (1 + 2 + 3 + \ldots + n) \approx 0.5(n^2)$

Average cost per operation is
$C(n)/n \approx 0.5n$


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# Using Amortized Analysis…

**Consider a sequence of operations with cost**

$$c_k = \begin{cases} 1 + 2^j & \text{if } (k-1) = 2^j \\ 1 & \text{otherwise} \end{cases}$$

***Worst case***

We will show ***later*** that

$C(n) \le 3n$ for all $n$

Amortized cost per operation = $C(n)/n = 3$

Amortized Time is Constant $\Theta(1)$


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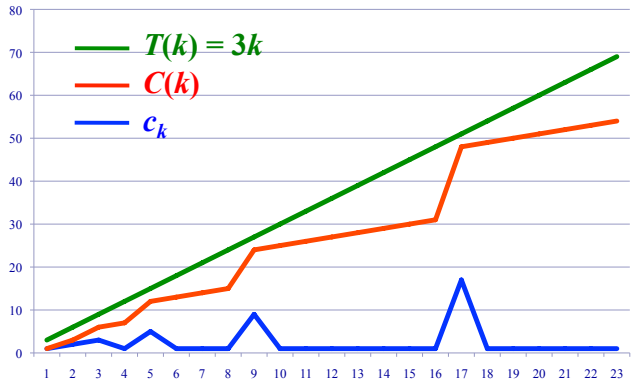
# Graphical demonstration

- **For now, I only give a graphical demo**





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# Summary: For this example

Worst-case bound over-estimates

- It give average $\leq 0.5n$

Amortized complexity analysis (*shown later*)

- Give average cost per operation $\leq 3$

Sometimes, the worst-case time of an operation does not accurately capture the worst-case time of a *sequence of operation*.


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# Different Complexity Analysis

- **Worst-Case Time:**
  - Worst-case of any operation $O_k$ over all possible instances. (Worst-case *actual time* of $c_k$ for any $k$).
- **Average-case Time:**
  - Average-case of each operation over all instances taken from certain probability distribution.
- **Amortized Time:**
  - Look at **worst-case sequence** of operations, and ask what is the *average cost per operation* for this worst-case sequence.


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# Motivation!

- Amortized analysis ***guarantees*** the average cost per operation in the ***worst-case sequence of operation***.
- Even though a ***bad single operation*** may be ***expensive***, the ***average cost*** may be small.
- Even in a worst-case sequence of operations, the ***worst-case performance does not happen very often***


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# A First Example: Binary Counter

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# Incrementing a Binary Counter

- ***k*-bit Binary Counter: *A*[0..*k*-1]**

$$x = \sum_{i=0}^{k-1} A[i] \cdot 2^i$$

```
INCREMENT(A)
i ← 0
while i < length[A] and A[i] = 1
    do A[i] ← 0          ▷ reset a bit
       i ← i + 1
if i < length[A]
    then A[i] ← 1        ▷ set a bit
```


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# *k*-bit Binary Counter

| Ctr | A[4] | A[3] | A[2] | A[1] | A[0] | $c_k$ | $C(k)$ |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | | |
| 1 | 0 | 0 | 0 | 0 | 1 | *1* | *1* |
| 2 | 0 | 0 | 0 | 1 | 0 | *2* | *3* |
| 3 | 0 | 0 | 0 | 1 | 1 | *1* | *4* |
| 4 | 0 | 0 | 1 | 0 | 0 | *3* | *7* |
| 5 | 0 | 0 | 1 | 0 | 1 | *1* | *8* |
| 6 | 0 | 0 | 1 | 1 | 0 | *2* | *10* |
| 7 | 0 | 0 | 1 | 1 | 1 | *1* | *11* |
| 8 | 0 | 1 | 0 | 0 | 0 | *4* | *15* |
| 9 | 0 | 1 | 0 | 0 | 1 | *1* | *16* |
| 10 | 0 | 1 | 0 | 1 | 0 | *2* | *18* |
| 11 | 0 | 1 | 0 | 1 | 1 | *1* | *19* |
| 12 | 0 | 1 | 1 | 0 | 0 | *3* | *22* |

Let $c_k$ be the # of bits flipped during the $k^{th}$ INCREMENT

And $C(k)$ be the $(c_1 + c_2 + \ldots + c_k)$


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# Worst-case analysis

Consider a sequence of $n$ increments.
In **worst-case**, number of bit flipped during increment is $k$. Thus, worst-case *total time* for $n$ insertions is $C(n) \le n \cdot k = O(n \cdot k)$.

But it is **WRONG** to say that $C(n) = \Theta(n \cdot k)$.

In fact, the worst-case cost for $n$ insertions is only $\Theta(n) \ll \Theta(n \cdot k)$.

Let's see why.


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# Tighter analysis

| Ctr | A[4] | A[3] | A[2] | A[1] | A[0] | $c_k$ | $C(k)$ |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | | 0 |
| 1 | 0 | 0 | 0 | 0 | 1 | 1 | 1 |
| 2 | 0 | 0 | 0 | 1 | 0 | 2 | 3 |
| 3 | 0 | 0 | 0 | 1 | 1 | 1 | 4 |
| 4 | 0 | 0 | 1 | 0 | 0 | 3 | 7 |
| 5 | 0 | 0 | 1 | 0 | 1 | 1 | 8 |
| 6 | 0 | 0 | 1 | 1 | 0 | 2 | 10 |
| 7 | 0 | 0 | 1 | 1 | 1 | 1 | 11 |
| 8 | 0 | 1 | 0 | 0 | 0 | 4 | 15 |
| 9 | 0 | 1 | 0 | 0 | 1 | 1 | 16 |
| 10 | 0 | 1 | 0 | 1 | 0 | 2 | 18 |
| 11 | 0 | 1 | 0 | 1 | 1 | 1 | 19 |
| 12 | 0 | 1 | 1 | 0 | 0 | 3 | 22 |

Total cost of $n$ operations

| | |
|---|---|
| A[0] flipped every op | $n$ |
| A[1] flipped every 2 ops | $n/2$ |
| A[2] flipped every 4 ops | $n/2^2$ |
| A[3] flipped every 8 ops | $n/2^3$ |
| … … … … | … |
| A[$i$] flipped every $2^i$ ops | $n/2^i$ |


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# Tighter analysis (continued)

Cost of $n$ increments $= \sum_{i=1}^{\lfloor \lg n \rfloor} \left\lfloor \frac{n}{2^i} \right\rfloor$

$< n \sum_{i=1}^{\infty} \frac{1}{2^i} = 2n$

$= \Theta(n)$

**Wow!**
Only 2 bits flipped per increment operation (*amortized*).

Thus, the average cost of each increment operation is $\Theta(n)/n = \Theta(1)$.

More precisely, the average cost of each increment operation is $\leq 2n/n = 2$


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# Look-back: *After* we know…

| Ctr | A[4] | A[3] | A[2] | A[1] | A[0] | $c_k$ | $C(k)$ |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | | 0 |
| 1 | 0 | 0 | 0 | 0 | 1 | 1 | 1 |
| 2 | 0 | 0 | 0 | 1 | 0 | 2 | 3 |
| 3 | 0 | 0 | 0 | 1 | 1 | 1 | 4 |
| 4 | 0 | 0 | 1 | 0 | 0 | 3 | 7 |
| 5 | 0 | 0 | 1 | 0 | 1 | 1 | 8 |
| 6 | 0 | 0 | 1 | 1 | 0 | 2 | 10 |
| 7 | 0 | 0 | 1 | 1 | 1 | 1 | 11 |
| 8 | 0 | 1 | 0 | 0 | 0 | 4 | 15 |
| 9 | 0 | 1 | 0 | 0 | 1 | 1 | 16 |
| 10 | 0 | 1 | 0 | 1 | 0 | 2 | 18 |
| 11 | 0 | 1 | 0 | 1 | 1 | 1 | 19 |
| 12 | 0 | 1 | 1 | 0 | 0 | 3 | 22 |

**We know:**
**2** bits flipped per increment

**Savings:** (for future operations)

**Expensive ops:** (use savings)


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# A-problem: (for those interested)

**Did you see the pattern?**

$Q(1) = (1)$

$Q(2) = (1\ 2\ 1) = (1)\ 2\ (1)$

$Q(3) = (1\ 2\ 1\ 3\ 1\ 2\ 1) = (1\ 2\ 1)\ 3\ (1\ 2\ 1)$

$Q(4) = Q(3)\ 4\ Q(3)$

Solve for $Q(k)$


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# Amortized analysis

An ***amortized analysis*** is any strategy for analyzing a sequence of operations to show that the average cost per operation is small, even though a single operation within the sequence might be expensive.

- An amortized analysis *guarantees* the *average performance* of each operation in the *worst case*.


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# Types of amortized analyses

Three common amortization arguments:

- the ***aggregate*** method,
- the ***accounting*** method,
- ~~the ***potential*** method.~~

Will not cover potential method in Spring 2015 Sem

We've just seen an aggregate analysis.

The aggregate method, though simple, lacks the precision of the other two methods. In particular, the accounting and potential methods allow a specific ***amortized cost*** to be allocated to each operation.


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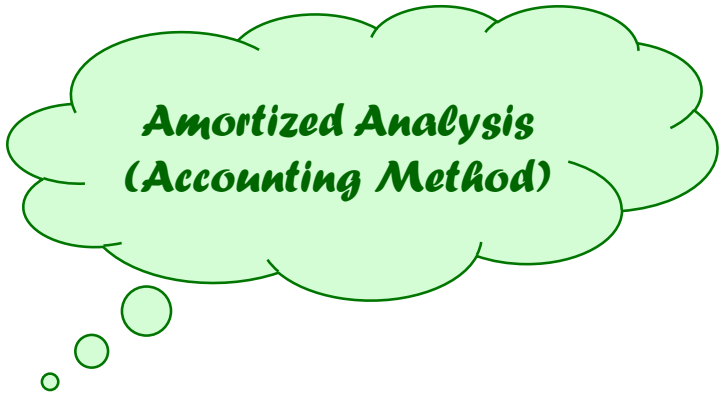
# A First Example: Binary Counter





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# Accounting method

- Charge $i$th operation a fictitious ***amortized cost*** $\hat{c}_i$, where $1 pays for 1 unit of work (*i.e.*, time).
- This fee is consumed to perform the operation.
- Any amount not immediately consumed is stored in the ***bank*** for use by subsequent operations.
- The bank balance must not go negative! We must ensure that

$$\sum_{i=1}^{n} c_i \le \sum_{i=1}^{n} \hat{c}_i$$

for all $n$.
- Then, the total amortized costs provide an upper bound on the total true costs.


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# Accounting analysis of INCREMENT

Charge an amortized cost of $2 every time a bit is set from 0 to 1

- $1 pays for the actual bit setting.
- $1 is stored for later re-setting (from 1 to 0).

At any point, every 1 bit in the counter has $1 on it… that pays for resetting it. (reset is "*free*")

**Example:**

$0\ 0\ 0\ 1^{\$1}\ 0\ 1^{\$1}\ 0$

$0\ 0\ 0\ 1^{\$1}\ 0\ 1^{\$1}\ 1^{\$1}$ **Cost = $2**

$0\ 0\ 0\ 1^{\$1}\ 1^{\$1}\ 0\ 0$ **Cost = $2**


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# Incrementing a Binary Counter

```
INCREMENT(A)
1.  i ← 0
2.  while i < length[A] and A[i] = 1
3.      do A[i] ← 0          ▷ reset a bit
4.         i ← i + 1
5.  if i < length[A]
6.      then A[i] ← 1        ▷ set a bit
```

- **When Incrementing,**
  - **Amortized cost for line 3 = $0** // use savings
  - **Amortized cost for line 6 = $2** // cost for setting bit
- **Amortized cost for INCREMENT($A$) = $2**
- **Amortized cost for $n$ INCREMENT($A$) = $2$n$ = $O(n)$**


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# Accounting analysis (continued)

**Key invariant:** Bank balance never drops below 0. Thus, the sum of the amortized costs provides an upper bound on the sum of the true costs.

| $i$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| #1's | 1 | 1 | 2 | 1 | 2 | 2 | 3 | 1 | 2 | 2 |
| $c_i$ | 1 | 2 | 1 | 3 | 1 | 2 | 1 | 4 | 1 | 2 |
| $\hat{c}_i$ | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| $bank_i$ | 1 | 1 | 2 | 1 | 2 | 2 | 3 | 1 | 2 | 2 |


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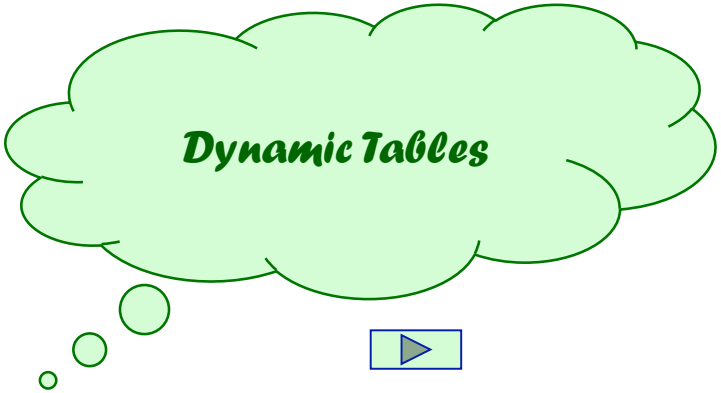
# Second Example:





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**Thank you.**


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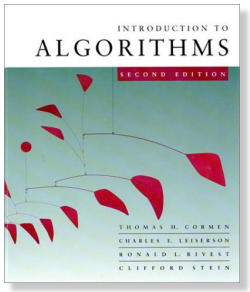
# *Introduction to Algorithms*

## 6.046J/18.401J




**LECTURE 6**

**Amortized Analysis**

- Dynamic tables
- Aggregate method
- Accounting method
- ~~Potential method~~

**Prof. Charles E. Leiserson**

With additional slides added by LeongHW (2013-2014)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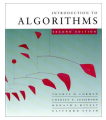


# Example 2: Dynamic Table

Slide added by LeongHW (Nov 2013)

# How large should a hash table be?

**Goal:** Make the table as small as possible, but large enough so that it won't overflow (or otherwise become inefficient).

**Problem:** What if we don't know the proper size in advance?

**Solution:** ***Dynamic tables.***

**IDEA:** Whenever the table overflows, "grow" it by allocating (via **`malloc`** or **`new`**) a new, larger table. Move all items from the old table into the new one, and free the storage for the old table.



# Dynamic Arrays

**Used in:** Java ArrayList and C++ Vector

**Check its API for Java ArrayList:**
http://www.docjar.com/html/api/java/util/ArrayList.java.html

especially API and code for methods
add, ensureCapacityInternal, grow

We illustrate with this

**Check its API for C++ Vector:**
http://www.cplusplus.com/reference/vector/vector/

especially API and code for methods
push_back, insert, resize

Slide added by LeongHW (Nov 2013)



# ArrayList API and impl (1)

**Code for add**

```
    /**
     * Appends the specified element to the end of this list.
     *
     * @param e element to be appended to this list
     * @return <tt>true</tt> (as specified by {@link Collection#add})
     */
    public boolean add(E e) {
        ensureCapacityInternal(size + 1);  // Increments modCount!!
        elementData[size++] = e;
        return true;
    }
```

Slide added by LeongHW (Nov 2013)



# ArrayList API and impl (2)

**Code for ensureCapacityInternal**

```
    /**
     * Increases the capacity of this <tt>ArrayList</tt> instance, if
     * necessary, to ensure that it can hold at least the number of elements
     * specified by the minimum capacity argument.
     *
     * @param   minCapacity   the desired minimum capacity
     */
    public void ensureCapacity(int minCapacity) {
        if (minCapacity > 0)
            ensureCapacityInternal(minCapacity);
    }

    private void ensureCapacityInternal(int minCapacity) {
        modCount++;
        // overflow-conscious code
        if (minCapacity - elementData.length > 0)
            grow(minCapacity);
    }
```

Slide added by LeongHW (Nov 2013)

# ArrayList API and impl (3)

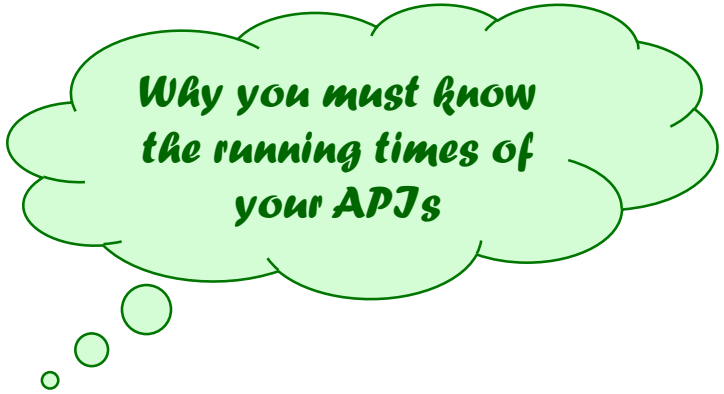
## Code for grow

```
198     /**
199      * Increases the capacity to ensure that it can hold at least the
200      * number of elements specified by the minimum capacity argument.
201      *
202      * @param minCapacity the desired minimum capacity
203      */
204     private void grow(int minCapacity) {
205         // overflow-conscious code
206         int oldCapacity = elementData.length;
207         int newCapacity = oldCapacity + (oldCapacity >> 1);
208         if (newCapacity - minCapacity < 0)
209             newCapacity = minCapacity;
210         if (newCapacity - MAX_ARRAY_SIZE > 0)
211             newCapacity = hugeCapacity(minCapacity);
212         // minCapacity is usually close to size, so this is a win:
213         elementData = Arrays.copyOf(elementData, newCapacity);
214     }
```

Slide added by LeongHW (Nov 2013)



# Side-Track:





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# Important to know your APIs well

www.docjar.com/docs/api/java/util/ArrayList.html

Home » openjdk-7 » java » util » [javadoc | source]

java.util

**public class: ArrayList** [javadoc | source]

java.lang.Object
└ java.util.AbstractCollection<E>
└ java.util.AbstractList<E>
└ **java.util.ArrayList**

**All Implemented Interfaces:**
Cloneable, List, Serializable, RandomAccess, Collection

**Direct Known Subclasses:**
RoleList, AttributeList, RoleUnresolvedList

Resizable-array implementation of the `List` interface. Implements all optional list operations, and permits all elements, including `null`. In addition to implementing the `List` interface, this class provides methods to manipulate the size of the array that is used internally to store the list. (This class is roughly equivalent to `Vector`, except that it is unsynchronized.)

The `size`, `isEmpty`, `get`, `set`, `iterator`, and `listIterator` operations run in constant time. The `add` operation runs in *amortized constant time*, that is, adding n elements requires O(n) time. All of the other operations run in linear time (roughly speaking). The constant factor is low compared to that for the `LinkedList` implementation.

http://www.docjar.com/docs/api/java/util/ArrayList.html



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www.docjar.com/docs/api/java/util/ArrayList.html

Home » openjdk-7 » java » util » [javadoc | source]

java.util

**public class: ArrayList** [javadoc | source]

java.lang.Object
└ java.util.AbstractCollection<E>
└ java.util.AbstractList<E>
└ **java.util.ArrayList**

**All Implemented Interfaces:**
Cloneable, List, Serializable, RandomAccess,

**Direct Known Subclasses:**
RoleList, AttributeList, RoleUnresolvedList

Resizable-array implementation of the `List` interface. Implements all optional list operations, and permits all elements, including `null`. In addition to implementing the `List` interface, this class provides methods to manipulate the size of the array that is used internally to store the list. (This class is roughly equivalent to `Vector`, except that it is unsynchronized.)

The `size`, `isEmpty`, `get`, `set`, `iterator`, and `listIterator` operations run in constant time. The `add` operation runs in *amortized constant time*, that is, adding n elements requires O(n) time. All of the other operations run in linear time (roughly speaking). The constant factor is low compared to that for the `LinkedList` implementation.

Bad. . .

All *other* operations run in linear time.

"Array expansion" is *amortized constant time*.

Good


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# Wrong use of API (unwittingly)

- In Spr-2014 PA1-ABC, you used `ArrayList`
  - To get the rank (after sorting), you call the API: `indexOf()`
  - To check for a student, you call the API: `contains()`

**What's Good:**
Easy and fast to do!
Done in no time at all.

**What's FATAL:**
TLE for task C
too slow (linear search)!

**Solution:** Code a O(lg *n*) binary search API


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# Moral of the Story

**"Theorem":**
Just because the API is available in the library
(& it's so easy to call it – just one loc)
does NOT mean you should always call it.

Especially, if you want your code to *scale*.

It is important to know your APIs well.
Remember *Analysis*! Analyze and know the running times of the APIs before you use them.


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# Careful with Online Help

**Not all online help gives running time.**

**Official doc for Java ArrayList:**
* http://www.docjar.com/docs/api/java/util/ArrayList.html

Good! Gives running times of all APIs.

**Some help resources on Java ArrayList:**
* http://developer.android.com/reference/java/util/ArrayList.html
* http://www.tutorialspoint.com/java/java_arraylist_class.htm

Don't show running times of API (not their focus)
Good for novice and beginners;


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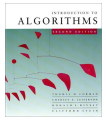
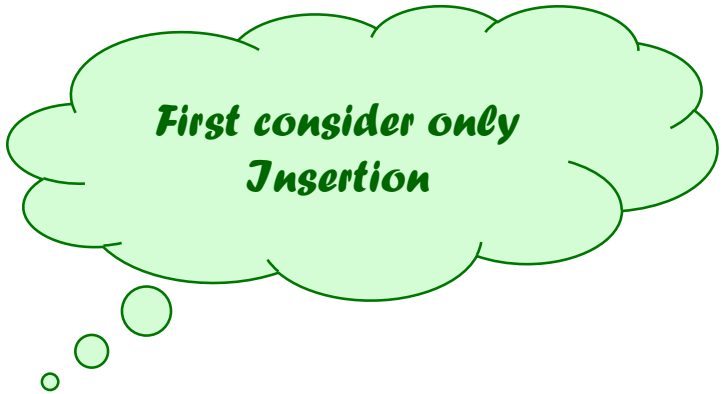


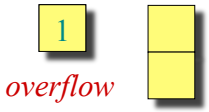

# Dynamic Tables




Slide added by LeongHW (Nov 2013)

# Example of a dynamic table

1. INSERT
2. INSERT

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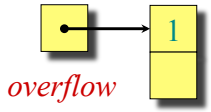
# Example of a dynamic table

1. INSERT
2. INSERT

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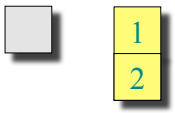
1. INSERT
2. INSERT

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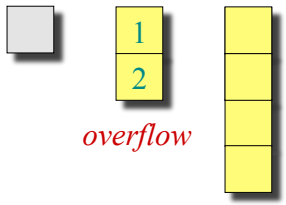
# Example of a dynamic table

1. INSERT
2. INSERT
3. INSERT

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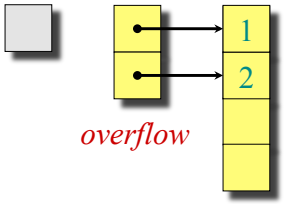
2. INSERT
3. INSERT

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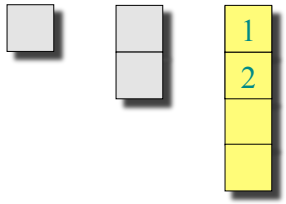
1. INSERT
2. INSERT
3. INSERT

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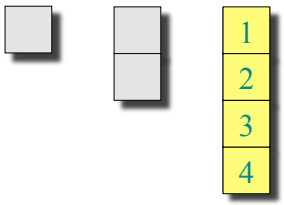
1. INSERT
2. INSERT
3. INSERT
4. INSERT

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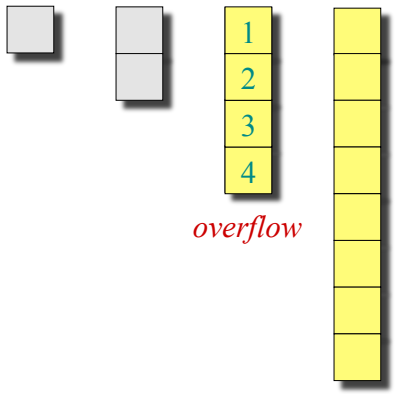
# Example of a dynamic table

1. INSERT
2. INSERT
3. INSERT
4. INSERT
5. INSERT

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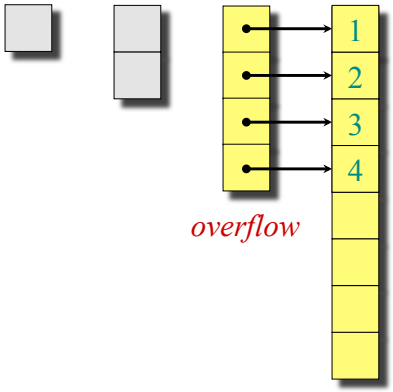


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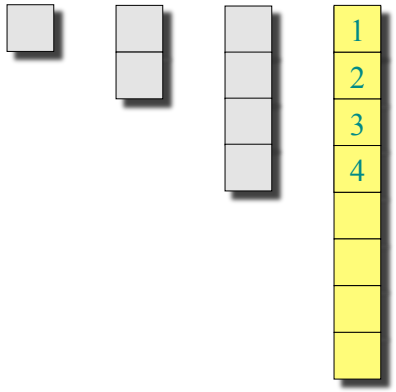


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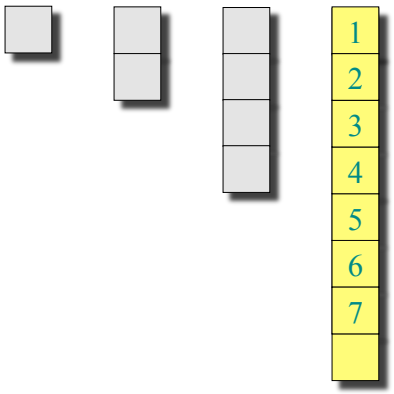
# Example of a dynamic table

1. INSERT
2. INSERT
3. INSERT
4. INSERT
5. INSERT
6. INSERT
7. INSERT

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# Worst-case analysis

Consider a sequence of $n$ insertions. The worst-case time to execute one insertion is $\Theta(n)$. Therefore, the worst-case time for $n$ insertions is $n \cdot \Theta(n) = \Theta(n^2)$.

**WRONG!** In fact, the worst-case cost for $n$ insertions is only $\Theta(n) \ll \Theta(n^2)$.

Let's see why.

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# Dynamic Tables (Insert)

**Amortized Analysis: Aggregate Method**

Slide added by LeongHW (Nov 2013)

# Tighter analysis

Let $c_i =$ the cost of the $i$th insertion

$$= \begin{cases} i & \text{if } i-1 \text{ is an exact power of } 2, \\ 1 & \text{otherwise.} \end{cases}$$

| $i$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $size_i$ | 1 | 2 | 4 | 4 | 8 | 8 | 8 | 8 | 16 | 16 |
| $c_i$ | 1 | 2 | 3 | 1 | 5 | 1 | 1 | 1 | 9 | 1 |

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# Tighter analysis

Let $c_i =$ the cost of the $i$th insertion

$$= \begin{cases} i & \text{if } i-1 \text{ is an exact power of } 2, \\ 1 & \text{otherwise.} \end{cases}$$

| $i$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $size_i$ | 1 | 2 | 4 | 4 | 8 | 8 | 8 | 8 | 16 | 16 |
| $c_i$ | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| | | 1 | 2 | | 4 | | | | 8 | |

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# Tighter analysis (continued)

$$\text{Cost of } n \text{ insertions} = \sum_{i=1}^{n} c_i$$

$$\le n + \sum_{j=0}^{\lfloor \lg(n-1) \rfloor} 2^j$$

$$\le 3n$$

$$= \Theta(n).$$

Thus, the average cost of each dynamic-table operation is $\Theta(n)/n = \Theta(1)$.

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# Recall from Introduction…

Slide added by LeongHW (Nov 2013)


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# Example: Consider this sequence

## Consider a sequence of operations with cost

| $k$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $c_k$ | 1 | 2 | 3 | 1 | 5 | 1 | 1 | 1 | 9 | 1 |

| $k$ | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|
| $c_k$ | 1 | 1 | 1 | 1 | 1 | 1 | 17 | 1 | 1 | 1 |

$$c_k = \begin{cases} k & \text{if } (k-1) \text{ is a power of 2} \\ 1 & \text{otherwise} \end{cases}$$

Slide added by LeongHW (Nov 2013)


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# Using Amortized Analysis…

## Consider a sequence of operations with cost

$$c_k = \begin{cases} 1 + 2^j & \text{if } (k-1) = 2^j \\ 1 & \text{otherwise} \end{cases}$$

*Worst case*

We will show ***later*** that

$C(n) \le 3n$ for all $n$

Amortized cost per operation = $C(n)/n = 3$

Amortized Time is Constant $\Theta(1)$


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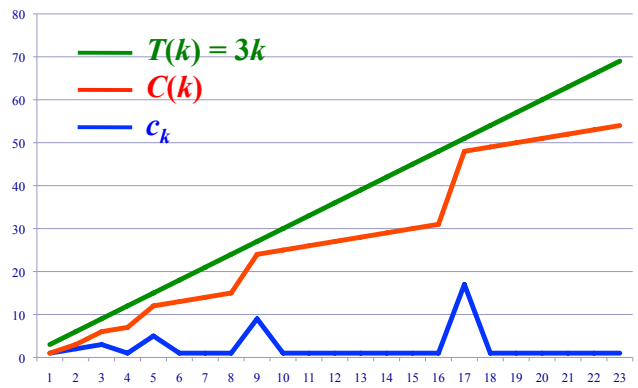
# Graphical demonstration

- For now, I only give a graphical demo




Slide added by LeongHW (Nov 2013)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# Dynamic Tables (Insert)

**Amortized Analysis: Accounting Method**

Slide added by LeongHW (Nov 2013)

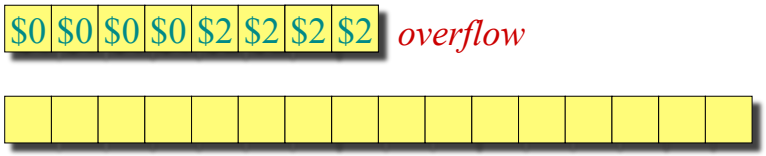
# Accounting analysis of dynamic tables

Charge an amortized cost of $\hat{c}_i = \$3$ for the $i$th insertion.

- $1 pays for the immediate insertion.
- $2 is stored for later table doubling.

When the table doubles, $1 pays to move a recent item, and $1 pays to move an old item.

**Example:**



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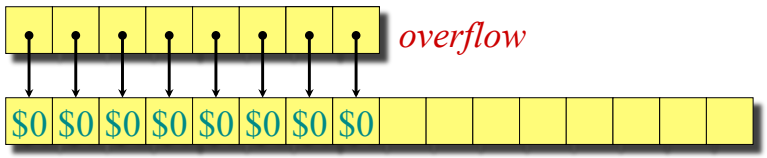




# Accounting analysis of dynamic tables

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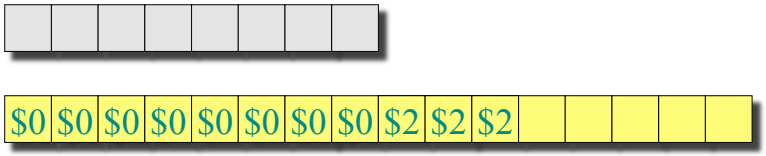
**Example:**



# Accounting analysis (continued)

**Key invariant:** Bank balance never drops below 0. Thus, the sum of the amortized costs provides an upper bound on the sum of the true costs.

| $i$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $size_i$ | 1 | 2 | 4 | 4 | 8 | 8 | 8 | 8 | 16 | 16 |
| $c_i$ | 1 | 2 | 3 | 1 | 5 | 1 | 1 | 1 | 9 | 1 |
| $\hat{c}_i$ | 2* | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| $bank_i$ | 1 | 2 | 2 | 4 | 2 | 4 | 6 | 8 | 2 | 4 |

*Okay, so I lied. The first operation costs only $2, not $3.

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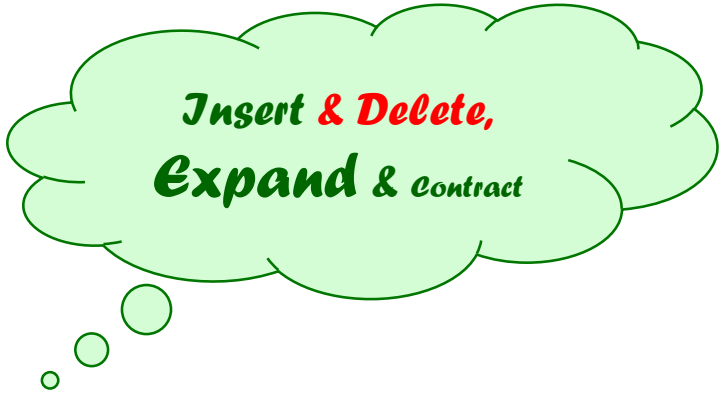
# Dynamic Tables




Slide added by LeongHW (Nov 2013)

# When to Contract the table?

**Solution 1: for Insert & Delete**

| | |
|---|---|
| Table overflows | Double the table size |
| Table < ½ full | Halve the table size |

**NO!** ***Bad*** idea! WHY?

Can cause trashing.

Table Size = 8; If we perform
I, D, D, I, I, D, D, I, I, D, D

Slide added by LeongHW (Nov 2013)

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# When to Contract the table?

**Solution 2: for Insert & Delete**

| | | |
|---|---|---|
| Table overflows | ➔ | Double the table size |
| Table < ¼ full | ➔ | Halve the table size |

This allows the table to save up some credit when $\alpha(T)$ is between ½ to ¼
To pay for moving old items to contracted table

Define load factor $\alpha(T)$
$\alpha(T) = num[T] / size[T]$

Slide added by LeongHW (Nov 2013)

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# Insert & Delete : with Accounting Method

**No change to Insert operation**

**Insert:** amortized cost of $\hat{c}_i = \$3$ (no change)

- $1 pays for the immediate insertion.
- $2 is stored for later table doubling.

When the table doubles, $1 pays to move a recent item, and $1 pays to move an old item.

Slide added by LeongHW (Nov 2013)

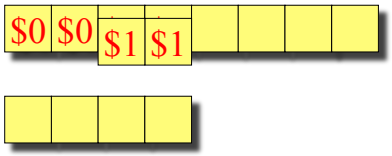
# Accounting: Insert & Delete

**Delete:** amortized cost of $\hat{c}_i = \$2$

- $1 pays for the immediate deletion.
- $1 stored in deleted cell, for later table halving.

When the table contracts, $1 pays to copy the old item into new contracted table.

**Example:**




*underflow*

Slide added by LeongHW (Nov 2013)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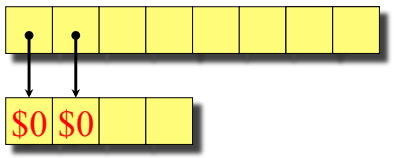
# Accounting: Insert & Delete

**Delete:** amortized cost of $\hat{c}_i = \$2$

- $1 pays for the immediate deletion.
- $1 stored in deleted cell, for later table halving.

When the table contracts, $1 pays to copy the old item into new contracted table.

**Example:**




*underflow*

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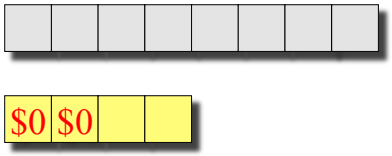


Slide added by LeongHW (Nov 2013)


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# Accounting: Insert & Delete

**Insert:** amortized cost of $\hat{c}_i = \$3$

**Delete:** amortized cost of $\hat{c}_i = \$2$

Therefore, $n$ insert/delete cost $\Theta(n)$ in worst-case.

Therefore, $n$ insert/delete cost $\leq 3n$ in worst-case.

Yes, we do "lose" *some* of the credits saved. (*Where*?) But, still amortized constant time.

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# Conclusions

- Amortized costs can provide a clean abstraction of data-structure performance.
- Any of the analysis methods can be used when an amortized analysis is called for, but each method has some situations where it is arguably the simplest or most precise.
- Different schemes may work for assigning amortized costs in the accounting method, or potentials in the potential method, sometimes yielding radically different bounds.

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